Developing Predictive Models for $PM_{2.5}$ Concentrations

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August 2025

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Abstract

Accurate, uncertainty-aware estimation of monthly site-level $PM_{2.5}$ is essential for surveillance, policy evaluation, and exposure assessment. This dissertation builds and evaluates a deployment-oriented pipeline for Scotland that compares three learners trained on $Z = \log(PM_{2.5} + 1)$: (i) a stepwise-AIC linear model (LM), (ii) an ℓ_1-penalised regression (LASSO), and (iii) a Random Forest (RF) paired with a Quantile Regression Forest (QRF). Predictions are back-transformed with Duan's smearing to reduce retransformation bias. Validation follows a site-aware design: leave-one-site-out (LOSO) within training and a strict 80/20 site-level hold-out with no site overlap. Performance is assessed by RMSE, MAE, R^2, and empirical coverage/width of prediction intervals on the original μg/m^3 scale. Distribution-free uncertainty is delivered via split- and cross-validated conformal prediction; QRF provides heteroskedastic conditional quantiles.

On the strict site hold-out, RF/QRF attains the strongest generalisation (RMSE = 1.43 μg/m^3, MAE = 0.963 μg/m^3, R^2 = 0.658) and the narrowest intervals (mean width = 4.22 μg/m^3), with coverage moving toward 95% after CV-conformal calibration (width $\approx$ 4.46 μg/m^3). LM/LASSO remain competitive but consistently less accurate, reflecting nonlinearities and interactions captured by RF. Interpretation from LM partial effects and RF variable importance is coherent with atmospheric mechanisms: rainfall and near-surface wind reduce concentrations (removal/ventilation); sea-level pressure and humidity increase them (stagnation/moisture); a background proxy is positive; greenness (NDVI) is protective; strong seasonality persists after adjustment. LM diagnostics motivate HC3 and cluster-robust inference, while conformal intervals avoid parametric assumptions. Sensitivity analyses (collinearity screens, λ choice in LASSO, conformal fold changes, RF seeds) confirm stability of conclusions.

Finally, the tuned QRF is used to impute missing responses, filling 1,893 of 4,560 rows with predictive bands while preserving distributional and seasonal structure, and releasing an imputed dataset for downstream use. The resulting pipeline is reproducible and ready for deployment, with clear paths for drift-aware maintenance and refined uncertainty calibration.

Keywords: $PM_{2.5}$; air quality; Random Forest; Quantile Regression Forest; conformal prediction; Duan smearing; leave-one-site-out cross-validation; uncertainty quantification; variable importance; missing data imputation; Scotland.

Acknowledgements

I would like to express my deepest gratitude to **Professor Duncan Lee**, my dissertation supervisor, for his patient guidance, incisive feedback, and constant encouragement throughout this project. His clarity of thought and high standards have shaped both the direction of this work and my approach to research.

I am equally grateful to my academic adviser, **Dr Craig Wilkie**, for his steady mentorship, practical advice, and kindness over the course of the programme. His support helped me navigate challenges, stay focused, and keep improving.

My heartfelt thanks go to my **family** for their enduring support, companionship, and love. Your trust and encouragement have been my source of strength from the beginning to the end of this journey. I wish to express my deepest appreciation to my beloved partner, **Victoria Xu**; her selfless love and dedication have helped me become a better man, and her companionship is a gift for which I will be forever grateful.

I also thank all the **lecturers and teaching staff** at the University of Glasgow—particularly within the School of Mathematics and Statistics—for a rigorous and inspiring learning experience during my MSc year. Your teaching broadened my horizons and equipped me with the tools needed to complete this dissertation.

Chapter 1

Introduction to the Problem

Context and Importance

Ambient air pollution is one of the leading global environmental risk factors for premature mortality. In the most recent pre-pandemic Global Burden of Disease accounting, the *State of Global Air 2020* report estimated that, in **2019**, air pollution (from all sources) contributed to **6.67 million** deaths worldwide, of which **4.14 million** were attributable specifically to long-term exposure to ambient $PM_{2.5}$ [15]. These figures underscore the central role of fine particulate matter in cardiopulmonary morbidity and mortality.

To protect population health, the World Health Organization [36] now recommend substantially tighter guideline values for $PM_{2.5}$ (annual average 5 $\mu g/m^3$ and 24-hour 15 $\mu g/m^3$), reflecting evidence that adverse effects occur at lower concentrations than previously recognised. Policymakers increasingly need spatially and temporally resolved estimates of $PM_{2.5}$ to track progress against these benchmarks and to design targeted interventions.

Scottish and UK context. Within the UK, the national Automatic Urban and Rural Network (AURN) forms the backbone of compliance and assessment monitoring, but fixed-site monitors are necessarily sparse relative to population and land area, especially outside major cities [31]. Emission inventories and chemical transport models are therefore complemented by national-scale mapping (e.g., the Pollution Climate Mapping framework) reported annually by the UK administrations [8]. For Scotland specifically, air quality policy is articulated through the *Cleaner Air for Scotland 2* strategy, which emphasises improved evidence, exposure reduction, and health protection across urban and rural settings [27]. These needs motivate robust statistical and machine-learning models that can bridge the gaps between monitors, quantify uncertainty, and support scenario analysis.

Existing Literature and Research Gaps

A wide range of statistical and machine-learning approaches has been applied to $PM_{2.5}$ prediction, including linear and generalized linear models, tree ensembles such as random forests, gradient boosting, and deep learning. Random forests [5] remain a strong baseline thanks to their flexibility, built-in nonlinearities, and resilience to collinearity. For uncertainty-aware prediction, *Quantile Regression Forests* (QRF) directly target conditional quantiles rather than the mean, yielding predictive intervals that adapt to

heteroskedasticity [21]. In parallel, *conformal prediction* provides *distribution-free* predictive coverage guarantees under minimal assumptions and can be layered on top of any point predictor, including forests and neural networks [16, 34]. Recent refinements such as conformalized quantile regression improve conditional coverage in finite samples while preserving efficiency [25].

Despite these advances, several gaps remain for regions like Scotland where (i) monitors are relatively sparse and heterogeneously distributed; (ii) exposure drivers span complex meteorology, land cover, population density, and proximity to traffic; and (iii) actionable assessments require *spatial generalisability* beyond the monitored network. Many published studies rely on random (non-spatial) cross-validation, which can overstate performance; rigorous spatial validation—e.g., site-based leave-one-location-out schemes—is essential to approximate real-world deployment. Moreover, uncertainty quantification is often limited to algorithm-specific intervals without formal coverage guarantees. Addressing these gaps calls for models that pair strong predictive accuracy with *valid*, *transparent* uncertainty statements.

Aims of the Proposed Research

This dissertation has two overarching aims:

1. To systematically benchmark statistical and machine-learning approaches for predicting *monthly* average $PM_{2.5}$ across Scotland, with a focus on methods that balance accuracy, interpretability, and scalability (linear models, penalised regression, random forests, and quantile forests).

2. To deliver *calibrated* uncertainty quantification for spatial predictions via split-conformal methods layered on random-forest predictors, enabling valid predictive intervals under realistic distributional shift.

Questions of interest

The main questions of interest are:

- What is the best statistical or machine learning model for predicting monthly average $PM_{2.5}$ concentrations?

- With how much accuracy can monthly average $PM_{2.5}$ concentrations be predicted?

Contributions and Significance

Methodologically, we combine quantile forests with split-conformal calibration to obtain *distribution-free* predictive intervals tailored to Scotland's monitoring footprint. Empirically, we implement spatially structured validation (leave-one-site-out and held-out stations) to provide realistic accuracy estimates. Substantively, our analysis yields interpretable importance rankings of environmental drivers and uncertainty-aware maps that can inform Scottish and UK air-quality policy in the context of the WHO 2021 guideline targets [36].

Chapter 2

Description of the Data

Study design and data sources

This study uses monthly average concentrations of fine particulate matter ($PM_{2.5}$, $\mu g/m^3$) measured at multiple monitoring stations across Scotland from January 2016 to December 2020. Stations span roadside and urban-background environments and are accompanied by spatial coordinates and a set of meteorological, land/cover, and demographic covariates.

The analytical CSV file (`Data - PM25.csv`) contains:

- **Year**, **Month**, and a constructed **Date** (15th of each month);
- **Site**, **Type** (local environment), **UR** (urban–rural class);
- **Easting**, **Northing** (British National Grid, metres);
- **PM25** (response): monthly mean $PM_{2.5}$;
- **PM25_modelled**: annual mean modelled $PM_{2.5}$ for the 1 km^2 grid cell containing the site, from the UK *Pollution Climate Mapping* (PCM) product [32];
- Meteorology/land/people: **Hurs** (relative humidity), **Psl** (sea-level pressure), **Rainfall**, **SfcWind** (near-surface wind), **Sun** (sunshine hours), **Tas** (air temperature), **NDVI** (vegetation greenness; classical normalised-difference index [26]), **PopDen** (population density), **nearest_road_distance**.

Data loading, structure, and missing values

Variables were parsed to appropriate types (e.g., `Year` as integer; `Month` as an ordered factor Jan–Dec; coordinates as numeric; `Date` set to the 15th of each month). A missing-value audit shows non-random gaps in $PM_{2.5}$ because not all sites were operating throughout the full 60 months.

Table 2.1: Missing-value summary in the raw file.

Variable(s)	Number of missing values
`PM25`	1,893
All others (0): `Year, Month, Site, Type, UR, Easting, Northing, PM25_modelled, Hurs, Psl, Rainfall, SfcWind, Sun, Tas, NDVI, PopDen, nearest_road_distance, Date`	0

Rows with missing values in the response (`PM25`) or other essential predictors were removed to ensure transparent and reproducible modelling. This yielded a fully observed analysis set of **2,667** rows (row audit: raw $n = 4{,}560 \rightarrow$ after `drop_na` $n = 2{,}667$).

Pre-processing decisions and rationale

Collapsing rare site types. Very infrequent factor levels can destabilise estimation and inflate uncertainty. We therefore collapsed `Type` by keeping the three most prevalent levels and grouping the remainder as "Other". This preserves dominant contrasts while avoiding quasi-separation.

Standardisation of continuous covariates. For comparability and numerical stability in regression/regularisation, continuous predictors were z-standardised (mean 0, SD 1). The standardised variables are: `Hurs, Psl, Rainfall, SfcWind, Sun, Tas, NDVI, PopDen, nearest_road_distance, PM25_modelled, Easting, Northing`.

Exploratory data analysis (EDA)

We summarise dominant temporal, spatial, and linear-association patterns prior to modelling.

Graphical summaries

Temporal variation. Monthly and yearly boxplots (Figure 2.1) highlight higher concentrations in winter/early spring (Dec–Apr) and comparatively lower values in summer (Jul–Sep). A reduction is visible in 2020.

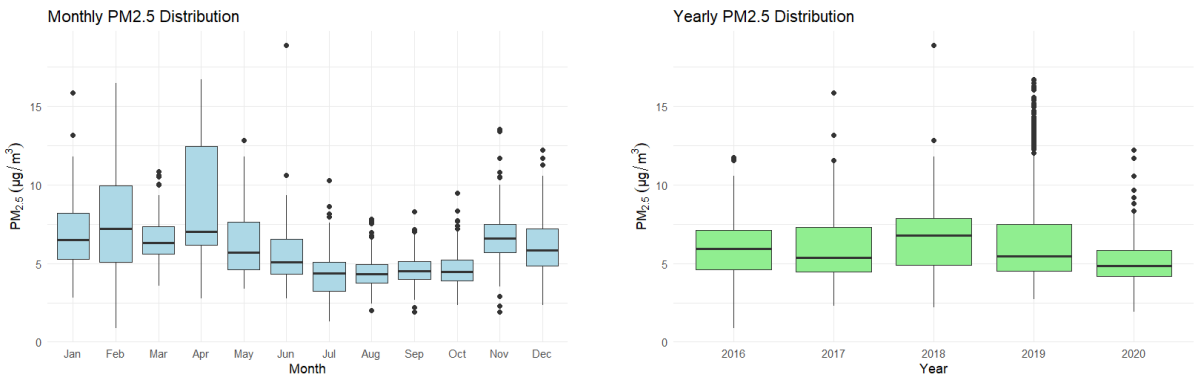


Figure 2.1: Monthly (left) and yearly (right) distributions of $PM_{2.5}$.

Spatial structure. Site-mean $PM_{2.5}$ (Figure 2.2) suggests higher values along urbanised corridors, indicating that population and road-proximity variables may aid prediction.

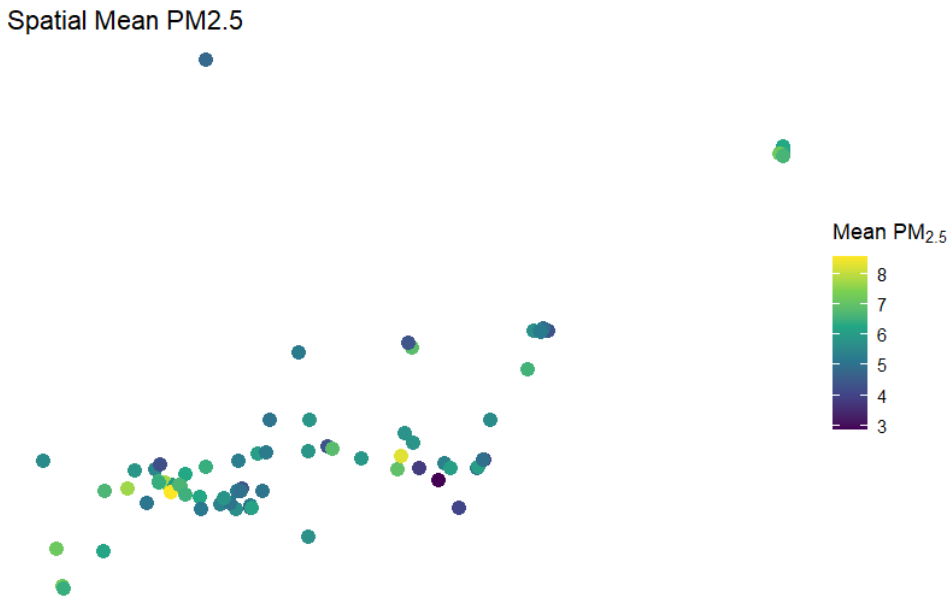


Figure 2.2: Spatial distribution of site-mean $PM_{2.5}$ across Scotland (points coloured by mean).

Correlation heatmap (visual). A heatmap of Pearson correlations (Figure 2.3) offers a quick view of associations and potential collinearity (e.g., humidity vs. sunshine; map axes). Numerical correlations are reported in Table 2.3.

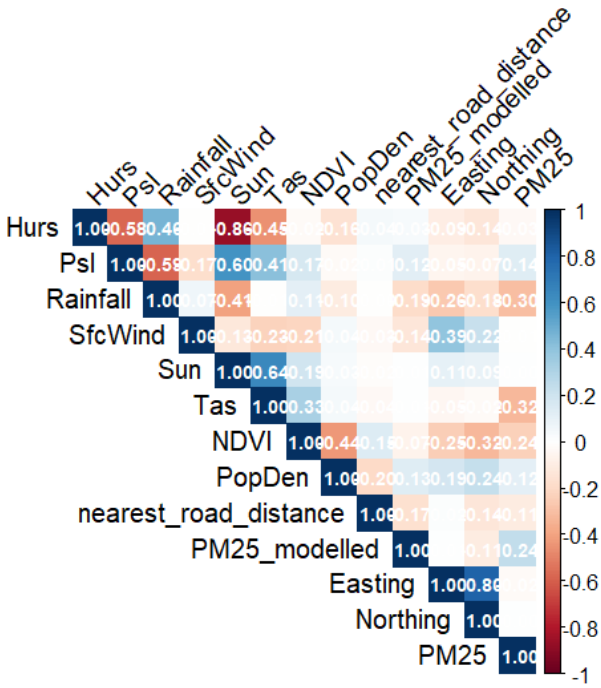


Figure 2.3: Correlation heatmap for $PM_{2.5}$ and continuous covariates (standardised).

Rationale and implementation of the log transformation

Environmental concentration data are typically right-skewed and close to log-normal [17, 23]. Fitting linear models on the raw scale induces right-tail leverage and strong mean–variance coupling (heteroskedasticity), weakening the normal-error working assumption for classical inference. A logarithmic transform is a standard remedy: it reduces skewness, stabilises variance, and makes multiplicative effects additive [4, 7, 22]. In Box–Cox terms, the log case ($\lambda \to 0$) usually yields residuals that are nearer homoskedastic and approximately normal, strengthening the validity of Wald-type SEs and parametric PIs when used. While OLS point estimates do not require strict normality, the log scale typically delivers more reliable residual behaviour in practice.

Empirical motivation in our data. For this dataset, the raw $PM_{2.5}$ distribution is right-skewed (sample skewness $= 1.521$), whereas $\log(PM_{2.5} + 1)$ is much closer to symmetric (skewness $= 0.435$). This contrast is visible in Figure 2.4. In addition, the residual histogram from a baseline log-scale model (Figure 2.5) is approximately symmetric, consistent with variance stabilisation on the log scale. These EDA results provide direct data-driven justification for modelling on a log scale in Chapter 3.

Transformation used and interpretation. We use

$$Y^{(\log)} = \log(PM_{2.5} + 1),$$

where the small constant protects zeros and is negligible for typical values. On this scale, an additive coefficient β_k corresponds to an approximate percentage change of $100[\exp(\beta_k) - 1]\%$ in $PM_{2.5}$ per one-unit increase in the covariate. Model fitting and formal diagnostics are conducted in Chapter 3; predictions are back-transformed via $\widehat{PM}_{2.5} = \exp(\widehat{Y}^{(\log)}) - 1$ for reporting.

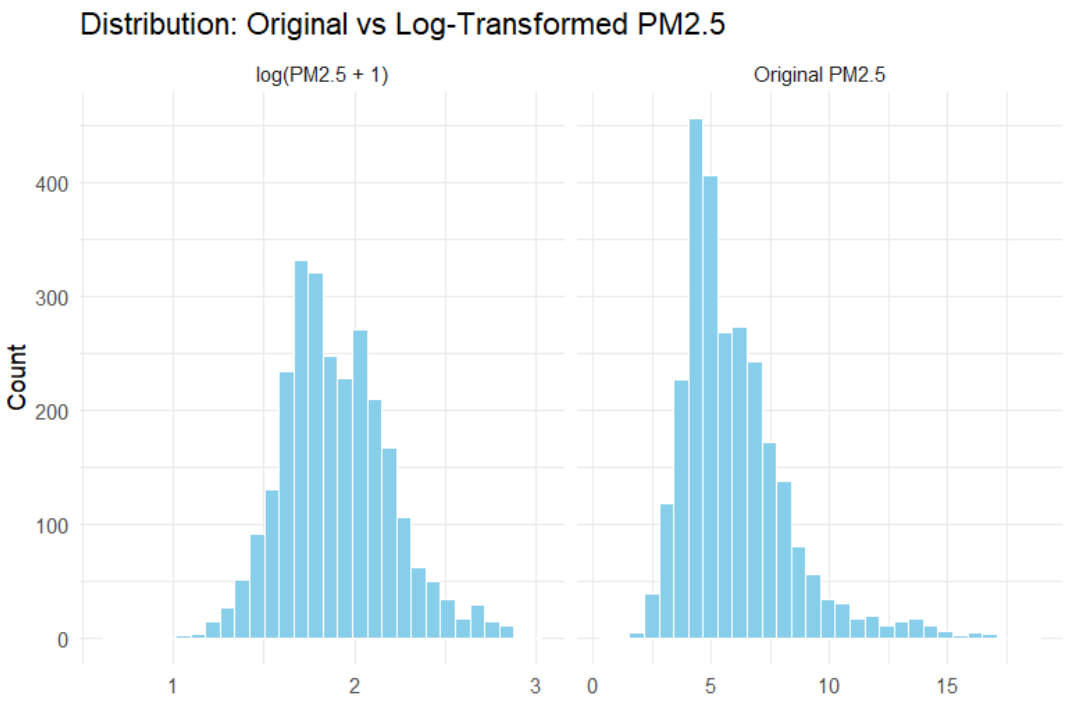


Figure 2.4: Distribution of $PM_{2.5}$ on the original scale (right) versus $\log(PM_{2.5} + 1)$ (left). The log scale is more symmetric.

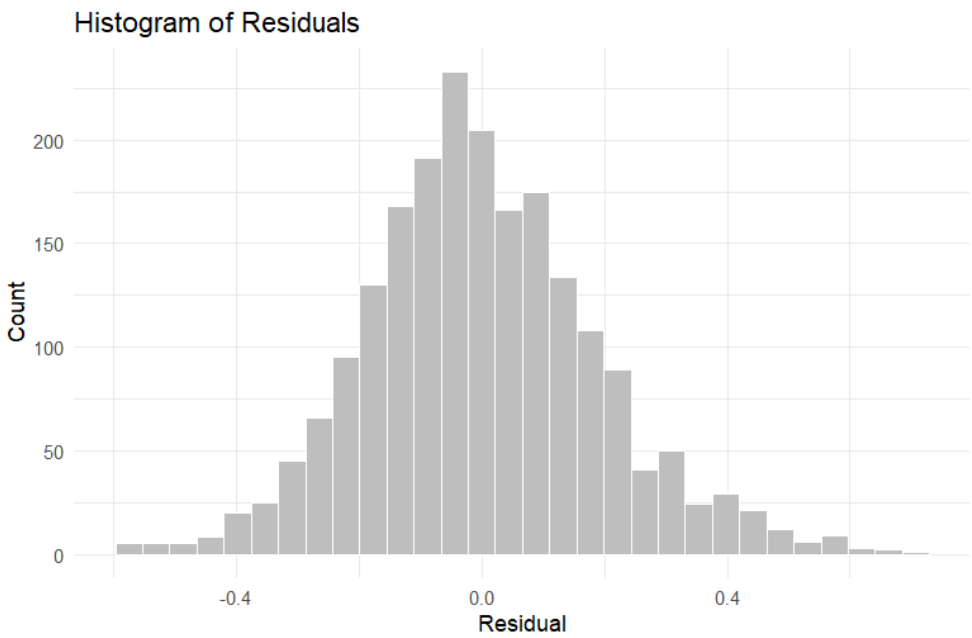


Figure 2.5: Residual histogram from an illustrative log-scale model: shape is close to normal, supporting variance stabilisation.

Numerical summaries

$PM_{2.5}$ by month and year

Tables 2.2a and 2.2b report the mean, SD, median and IQR on the original scale. Patterns align with the graphics: winter/early-spring peaks and a 2020 dip.

Table 2.2: Monthly and yearly summaries of $PM_{2.5}$ ($\mu g/m^3$).

(a) Monthly summary

Month	Mean	SD	Median	IQR
1	6.82	2.09	6.51	2.90
2	7.58	2.80	7.19	4.90
3	6.54	1.37	6.30	1.75
4	8.65	3.64	7.00	6.30
5	6.20	1.89	5.68	3.02
6	5.56	1.75	5.10	2.22
7	4.34	1.35	4.36	1.85
8	4.42	1.01	4.35	1.17
9	4.60	0.98	4.53	1.16
10	4.67	1.15	4.49	1.30
11	6.71	1.68	6.58	1.80
12	6.14	1.76	5.85	2.37

(b) Yearly summary

Year	Mean	SD	Median	IQR
2016	5.98	1.90	5.94	2.49
2017	5.99	2.00	5.38	2.83
2018	6.50	2.02	6.76	2.98
2019	6.69	3.09	5.48	2.97
2020	4.98	1.31	4.86	1.65

Correlation matrix (numeric)

Because covariates are standardised, the covariance matrix equals the correlation matrix. Table 2.3 reports Pearson correlations (pairwise complete observations) for continuous variables including $PM_{2.5}$. The strongest absolute correlations occur between meteorological variables (e.g., humidity vs. sunshine ≈ -0.86) and the map axes, motivating careful collinearity checks in Chapter 3.

Table 2.3: Correlation matrix (rounded to two decimals).

	Hurs	Psl	Rainfall	SfcWind	Sun	Tas	NDVI	PopDen	nearest_road_distance	PM25_modelled	Easting	Northing	PM25
Hurs	1.00	-0.58	0.46	-0.01	-0.86	-0.45	-0.02	-0.16	0.04	0.03	-0.09	-0.14	-0.03
Psl	-0.58	1.00	-0.59	-0.17	0.60	0.41	0.17	-0.02	0.01	0.12	-0.05	-0.07	0.14
Rainfall	0.46	-0.59	1.00	0.07	-0.41	-0.01	0.11	-0.10	0.00	-0.19	-0.26	-0.18	-0.30
SfcWind	-0.01	-0.17	0.07	1.00	-0.13	-0.23	-0.21	0.04	-0.03	-0.14	0.39	0.22	-0.01
Sun	-0.86	0.60	-0.41	-0.13	1.00	0.64	0.19	0.03	-0.02	0.01	0.11	0.09	0.00
Tas	-0.45	0.41	-0.01	-0.23	0.64	1.00	0.33	0.04	-0.04	0.01	-0.05	-0.02	-0.32
NDVI	-0.02	0.17	0.11	-0.21	0.19	0.33	1.00	-0.44	0.15	-0.07	-0.25	-0.32	-0.24
PopDen	-0.16	-0.02	-0.10	0.04	0.03	0.04	-0.44	1.00	-0.20	0.13	0.19	0.24	0.12
nearest_road_distance	0.04	0.01	0.00	-0.03	-0.02	-0.04	0.15	-0.20	1.00	-0.17	0.02	-0.14	-0.11
PM25_modelled	0.03	0.12	-0.19	-0.14	0.01	0.01	-0.07	0.13	-0.17	1.00	0.01	-0.11	0.24
Easting	-0.09	-0.05	-0.26	0.39	0.11	-0.05	-0.25	0.19	0.02	0.01	1.00	0.80	-0.02
Northing	-0.14	-0.07	-0.18	0.22	0.09	-0.02	-0.32	0.24	-0.14	-0.11	0.80	1.00	0.00
PM25	-0.03	0.14	-0.30	-0.01	0.00	-0.32	-0.24	0.12	-0.11	0.24	-0.02	0.00	1.00

Chapter 3

Description of the Methods

Overview and guiding principles

We model monthly site-level $PM_{2.5}$ in Scotland on the log scale, using three complementary learners: (i) a linear model (LM) with main effects, (ii) an ℓ_1-penalised regression (LASSO), and (iii) a nonparametric Random Forest (RF). LM provides interpretable elasticities and classical inference; LASSO performs data-adaptive selection under collinearity; RF captures nonlinearities and interactions without pre-specification [5, 13, 30, 37]. All models are trained on $\log(PM_{2.5} + 1)$ and predictions are back-transformed using Duan's smearing to reduce retransformation bias [9]. The log transform is also supported by the common empirical lognormality of pollutant concentrations and by the Box–Cox family, offering variance stabilisation and multiplicative-error interpretability [4, 7, 17, 23].

Out-of-sample performance is evaluated with *leave-one-site-out* (LOSO) on spatio-temporal data, and with a strict hold-out split by sites (80% train, 20% test) to emulate deployment at unseen locations [24]. For uncertainty quantification, we use *split-conformal* prediction intervals (distribution-free, marginally valid) for LM/LASSO and quantile forests for RF. We optionally apply conformal calibration to align coverage across methods [16, 21, 34].

Data notation and design choices

Index sites by $i = 1, \ldots, N$ and months by $t = 1, \ldots, T_i$. Let the original-scale response be $Y_{it} = PM_{2.5}$ (μg/m^3) and the transformed response

$$Z_{it} = \log(Y_{it} + 1).$$

Let $\mathbf{x}_{it} \in \mathbb{R}^p$ be the standardised continuous covariates (meteorology, land cover, demography, background $PM_{2.5}$, coordinates), and $\mathbf{d}_{it}$ the one-hot dummies for factors `Month`, `UR`, `Type` (baseline levels as in the data). Following the Chapter 2 collinearity screen, we drop `Sun` and `Northing` due to $|r| > 0.8$ with `Hurs` and `Easting`, respectively, and verify acceptable VIFs thereafter [12].

Model specifications in mathematical notation

LM assumes an additive conditional mean with roughly homoskedastic, weakly dependent errors; we probe violations via HC3 and site–cluster–robust standard errors, and the Durbin–Watson and Breusch–Pagan tests. LASSO keeps this mean but adds ℓ_1 shrinkage for selection under collinearity. RF is fully nonparametric—no error-distribution assumptions—trading coefficient-level interpretability for flexibility (nonlinearities and interactions).

(M1) Linear model on the log scale

$$Z_{it} \;=\; \alpha \;+\; \mathbf{x}_{it}^{\top}\boldsymbol{\beta} \;+\; \mathbf{d}_{it}^{\top}\boldsymbol{\gamma} \;+\; \varepsilon_{it},$$

with $E(\varepsilon_{it} \mid \mathbf{x}_{it}, \mathbf{d}_{it}) = 0$. OLS is used for point estimation. Inference is based on HC3 and site-cluster-robust SEs to mitigate heteroskedasticity and within-site dependence [3, 19, 35]. Residual serial correlation and heteroskedasticity are checked via Durbin–Watson and Breusch–Pagan tests [6, 10].

Back-transformation. For original-scale predictions,

$$\widehat{Y}_{it} \;=\; \exp(\widehat{Z}_{it}) \cdot \hat{S} - 1, \qquad \hat{S} = \frac{1}{n_{\mathcal{T}}} \sum_{(i,t)\in\mathcal{T}} \exp(\widehat{\varepsilon}_{it}),$$

where $\mathcal{T}$ indexes the training data used to compute the smearing factor.

(M2) LASSO (coordinate descent)

Let $\mathbf{z}$ be the stacked vector of Z_{it} and $\mathbf{X}$ the model matrix (intercept separated). LASSO solves

$$\widehat{\boldsymbol{\theta}}(\lambda) = \arg\min_{\boldsymbol{\theta}} \Big\{ \tfrac{1}{2n} \|\mathbf{z} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1 \Big\},$$

with $\lambda \geq 0$ tuned by cross-validation. We use `glmnet` with user-supplied LOSO folds; continuous predictors are standardised and factors are one-hot encoded [13, 30]. Back-transformation uses the same smearing adjustment.

Reproducibility. We report the one-standard-error choice λ_{1se} from LOSO-CV and fix this λ for all subsequent reporting.

(M3) Random Forest (RF) and quantile RF

RF averages B regression trees grown on bootstrap samples with feature subsampling m_{try} at each split:

$$\widehat{f}_{\text{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}).$$

Hyperparameters $(B, m_{\text{try}}, \text{min_node})$ are tuned by out-of-bag (OOB) error on the *log* target [5, 37]. For prediction intervals we use quantile regression forests (QRF) to estimate conditional quantiles of Z_{it} and then back-transform; optionally we apply a conformal correction on training residuals to align coverage across learners [16, 21]. For interpretation we report both impurity- and permutation-based variable importance [28, 29].

Reproducibility. We fix $(m_{\text{try}}, \text{min_node}, B)$ at the OOB-optimal values on the log target and reuse the same settings for RF (means) and QRF (quantiles), ensuring comparability and reproducibility.

Variable screening and collinearity control

We use AIC-based stepwise search (both directions) on the full LM to define a stable baseline formula (also used to build model matrices for LASSO/RF). AIC is AIC $= -2\log\mathcal{L} + 2k$ with k parameters [1]. Multicollinearity is mitigated by removing pairs with $|r| > 0.80$ and by inspecting variance inflation factors (VIFs); we target $\text{VIF}^{1/2} \lesssim 5$ as a pragmatic bound [12].

AIC-selected LM specification. Using bidirectional stepwise search by AIC over the final candidate set, we accepted moves only when they reduced AIC (by $\gtrsim 2$). The lowest-AIC specification—more parsimonious than the full model was:

$$\begin{aligned} \text{log_PM25} \sim\ & \texttt{factor(Month)} + \texttt{PM25_modelled} + \texttt{Tas} + \texttt{Rainfall} + \texttt{Hurs} \\ & + \texttt{Type} + \texttt{PopDen} + \texttt{SfcWind} + \texttt{UR} \\ & + \texttt{Psl} + \texttt{NDVI} + \texttt{nearest_road_distance}. \end{aligned} \tag{3.1}$$

Continuous covariates are z-standardised; factors are one-hot (January baseline for `Month`; most prevalent levels for `UR`/`Type`). Predictions are back-transformed with Duan smearing. We use Eq. (3.1) as the linear baseline in Chapters 3–4.

Validation design and uncertainty quantification

Train/test split by site and LOSO-CV

We split by *sites*: 80% of sites for training and 20% for a hold-out test (no site overlap). LOSO-CV holds out one site per fold (all its months) and trains on the rest, respecting spatial clustering and avoiding overly optimistic i.i.d. K-fold estimates [24].

Prediction intervals

For LM/LASSO we form *split-conformal* intervals on the original scale. With training predictions $\hat{y}_j$ and observations y_j $(j \in \mathcal{T})$,

$$q_\alpha = \text{Quantile}_{1-\alpha}(\,|y_j - \hat{y}_j| :\ j \in \mathcal{T}\,),$$

taken as the $\lceil (n_\mathcal{T}+1)(1-\alpha)\rceil$-th order statistic of absolute residuals. We report $[\hat{y} - q_\alpha,\ \hat{y} + q_\alpha]$ for new data, guaranteeing marginal coverage $\geq 1-\alpha$ under exchangeability [16, 34]. For RF we use QRF to obtain $\widehat{Q}_{\alpha/2}$ and $\widehat{Q}_{1-\alpha/2}$ for Z and then back-transform; optionally, the same q_α is applied for a conformal correction [21].

Performance metrics and definitions

For observed y_j, point predictions $\hat{y}_j$ $(j = 1, \ldots, n_{\text{eval}})$, and prediction intervals $[L_j, U_j]$:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(\hat{y}_j - y_j)^2}, \qquad \text{MAE} = \frac{1}{n}\sum_{j=1}^{n}|\hat{y}_j - y_j|,$$

$$R^2 = \Big[\text{corr}(\hat{y}, y)\Big]^2, \qquad \text{Coverage} = \frac{1}{n}\sum_{j=1}^{n}\mathbf{1}(L_j \le y_j \le U_j), \qquad \text{Width} = \frac{1}{n}\sum_{j=1}^{n}(U_j - L_j).$$

RMSE/MAE measure accuracy; R^2 summarises linear association. For non-linear learners (or specifications without an intercept), we report R^2 as $\Big[\text{corr}(\hat{y}, y)\Big]^2$ to ensure comparability across learners. Coverage and Width evaluate uncertainty quantification (high coverage with narrow width is desirable).

Diagnostics and robustness

Beyond global residual plots, we conduct:

1. Durbin–Watson test for residual autocorrelation [10].
2. Breusch–Pagan test for heteroskedasticity [6].
3. HC3 and site-cluster-robust standard errors for inference robustness [3, 19, 35].
4. Post-screening VIFs to ensure no harmful collinearity remains [12].

Overall, our inference relies on robust (HC3/cluster) SE to mitigate heteroskedasticity and within-site dependence; predictive assessment is entirely out-of-sample (LOSO and held-out sites); and interval coverage is guaranteed marginally by split-conformal methods without parametric assumptions.

Implementation details

All continuous covariates are standardised; factors are one-hot encoded; factor levels are harmonised across train/test. LASSO uses LOSO folds inside `cv.glmnet`. RF hyperparameters $(m_{\text{try}}, \text{min_node}, B)$ are tuned by OOB RMSE on the *log* target and then fixed. For comparable uncertainty across learners, we prefer conformal intervals with a common calibration quantile q_α computed on the training set. Numerical summaries in Chapter 2 are used only for set-up; formal training/CV/hold-out results are reported in Chapter 4. All randomisation uses fixed seeds; site-wise data splits are recorded for reproducibility.

Chapter 4

Analysis of the Data

Overview

This chapter evaluates predictive models for monthly site-level $PM_{2.5}$ in Scotland under a deployment-oriented design. All models are fitted to

$$Z = \log(PM_{2.5} + 1),$$

and predictions are back-transformed using Duan's smearing estimator to reduce retransformation bias [4, 9, 17, 23].

We benchmark three learners: (i) a stepwise-AIC linear model (LM) [1]; (ii) an ℓ_1-penalised regression (LASSO) via `glmnet` [13, 30]; and (iii) a Random Forest (RF) for means, paired with a Quantile Regression Forest (QRF) for conditional quantiles [5, 21, 37]. Evaluation uses a strict site-level 80/20 hold-out split (no site overlap) and leave-one-site-out (LOSO) cross-validation within training [24]. Primary metrics are RMSE, MAE, R^2, empirical coverage of nominal 95% prediction intervals (PIs), and mean PI width on the original $\mu g/m^3$ scale. Uncertainty is quantified via split- and CV-conformal prediction intervals with distribution-free finite-sample guarantees [2, 16, 25, 34].

Modelling setup recap

Relative to Chapter 3, two highly collinear covariates (`Sun`, `Northing`) were removed ($|r| > 0.8$); post-removal $\mathrm{GVIF}^{1/(2\,\mathrm{df})}$ values were acceptable. All continuous covariates are z-standardised; factors are one-hot with harmonised levels across train/test. We fit on $Z = \log(PM_{2.5} + 1)$ and back-transform with Duan smearing.

LM (AIC-selected). The linear baseline uses the AIC-selected specification given in Eq. (3.1) (Chapter 3), which balances fit and parsimony under correlated predictors.

LASSO. `cv.glmnet` with LOSO folds chose $\lambda_{1se} = 0.001334$, yielding a sparser, more stable model with essentially unchanged CV error vs. λ_{min}.

RF/QRF. Using `ranger`, we tuned by OOB RMSE on the log target and fixed $mtry$=12, `min.node.size`=5 (OOB $\mathrm{RMSE}_{log} = 0.135$) for both RF (means) and QRF (quantiles) to isolate loss-function effects. Seeds/versions are recorded for exact reproducibility.

Results

Predictive performance

Table 4.1 summarises accuracy and uncertainty across training (in-sample), LOSO, strict site hold-out, and CV-conformal hold-out. Lower RMSE/MAE/Width and higher R^2/Coverage are better.

Table 4.1: Performance across data splits (original scale, μg/m^3). Coverage is empirical proportion for nominal 95% PIs.

Model	Split	RMSE	MAE	R^2	Coverage	Width
LM	Training (in-sample)	1.55	1.10	0.540	0.951	6.27
LASSO	Training (in-sample)	1.63	1.13	0.504	0.951	6.65
RF/QRF	Training (in-sample)	0.263	0.076	0.988	0.997	4.22
LM	LOSO (train sites)	1.54	1.15	0.620	0.945	6.27
LASSO	LOSO (train sites)	1.58	1.15	0.582	0.949	6.67
RF/QRF	LOSO (train sites)	0.916	0.713	0.863	0.951	4.24
LM	Hold-out (unseen)	1.94	1.39	0.351	0.909	6.25
LASSO	Hold-out (unseen)	2.00	1.41	0.313	0.918	6.65
RF/QRF	Hold-out (unseen)	**1.43**	**0.963**	**0.658**	0.904	**4.22**
LM	Hold-out (CV-conformal)	1.94	1.39	0.351	0.936	6.85
LASSO	Hold-out (CV-conformal)	2.00	1.41	0.313	0.931	6.91
RF/QRF	Hold-out (CV-conformal)	1.43	0.963	0.658	0.914	4.46

Key findings.

- **Overall ranking at new sites.** On the strict hold-out, **RF/QRF** is the clear winner: RMSE = 1.43, MAE = 0.963, R^2=0.658, versus LM (1.94, 1.39, 0.351) and LASSO (2.00, 1.41, 0.313). This is a *26%* RMSE reduction vs. LM and *29%* vs. LASSO (MAE reductions *31%* and *32%*). Gains in R^2 (+0.307 vs. LM) indicate that forests capture important *nonlinearities and interactions*.

- **Generalisation gap.** Errors increase from training → LOSO → hold-out, as expected. Despite a larger flexibility-induced gap, RF still *dominates* where it matters—unseen sites—showing better spatial generalisability.

- **Interval efficiency.** For similar coverages around 91%, RF/QRF delivers *narrower* intervals (width = 4.22) than LM (6.25), a *32%* reduction. With CV-conformal calibration, RF coverage increases by +1.0 pp (to 91.4%) at a modest width increase to 4.46 (*+6%*). LM's coverage increases by +2.7 pp (to 93.6%) with a larger width penalty (+9.6%).

- **LOSO mirrors hold-out behaviour.** LOSO results anticipate the hold-out ordering and margins, indicating that the superiority of RF/QRF is not split-specific but reflects *spatial generalisability*.

Uncertainty quality

Conformal calibration nudges empirical coverage toward 95% with modest widening of PIs (e.g. RF/QRF width $4.22 \to 4.46$). QRF alone already adapts to heteroskedasticity; conformal post-calibration harmonises coverage across learners while preserving relative efficiency [16, 25].

Interpretation of fitted relationships

LM (log scale; Duan smearing $S = 1.0188$). Let $Z = \log(Y+1)$ and $\eta = \alpha + \mathbf{x}^\top \boldsymbol{\beta} + \mathbf{d}^\top \boldsymbol{\gamma}$. Continuous regressors are z–standardised, so $\partial\eta/\partial X_k = \beta_k$ and a one–SD increase in X_k induces a multiplicative change $\exp(\beta_k)$ in $\exp(\eta)$, hence an approximate percentage change $100\left(e^{\beta_k} - 1\right)\%$ in Y at fixed covariates (the global smearing factor S is multiplicative and does not affect these *comparative statics*). Factor effects are contrasts relative to the reference levels (January for `Month`; baseline levels for `UR`/`Type`).

Seasonality. Signs of the OLS-estimated coefficients for the monthly indicators are positive for February–April and negative for May–October relative to January, conditional on the meteorological and background covariates included in the model. This pattern indicates a residual seasonal component: higher winter/early-spring levels are consistent with increased domestic heating, shallower mixed layers, and occasional inversion episodes; lower summer coefficients accord with enhanced vertical mixing, more frequent wet scavenging, and more efficient photochemical removal of precursors.

Meteorology (partial effects).

- $\boldsymbol{\beta}_{\text{Hurs}} > 0$. Higher relative humidity is associated with increased aerosol water content (hygroscopic growth) and stagnant dispersion regimes, yielding $\partial\eta/\partial\text{Hurs} > 0$.
- $\boldsymbol{\beta}_{\text{Psl}} > 0$. Elevated sea-level pressure proxies anticyclonic stability and shallow boundary layers, again implying $\partial\eta/\partial\text{Psl} > 0$.
- $\boldsymbol{\beta}_{\text{Rainfall}} < 0$ and $\boldsymbol{\beta}_{\text{SfcWind}} < 0$. These comport with wet scavenging and mechanical ventilation, respectively.
- $\boldsymbol{\beta}_{\text{Tas}} > 0$ *conditional on* month/humidity/pressure, despite a negative marginal correlation. This is a textbook suppression/omitted-variable phenomenon: after conditioning on seasonal and stagnation structure, higher temperatures can coincide with episodes favourable to secondary formation or shoulder-season stability.

Background/setting. `PM25_modelled` (1 km background) enters positively as a regional baseline; NDVI enters negatively (greenness as a proxy for lower source density and enhanced deposition); non-urban UR levels are negative relative to the reference. The small positive coefficient for `nearest_road_distance` is not stable across folds and is plausibly confounded by `PopDen`/`Type` and sub-kilometre measurement error; we therefore refrain from substantive interpretation.

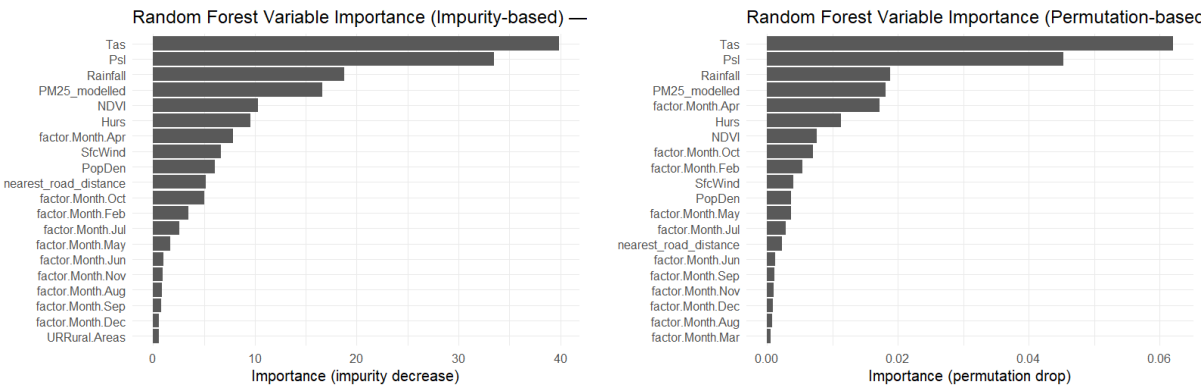
Inference limitations. Given tail heaviness, heteroskedasticity, and within-site dependence, coefficient uncertainty is summarised using HC3 and site–cluster–robust standard errors. All effects are *associational* rather than causal and pertain to the 2016–2020 monitoring footprint.

RF-based interpretation via variable importance. We compute importances on the log target Z using two criteria: (i) the *total node-size–weighted decrease in squared-error impurity* over all splits on a predictor, and (ii) the *out-of-bag permutation drop* in predictive accuracy, which mitigates impurity biases toward high-cardinality or early-splitting variables [28, 29]. We take permutation as the primary ranking and use impurity as a stability check. Importances are relative (unitless); with correlated drivers (e.g. *Tas*, *Hurs*, *Psl*, `Month`), signal can be *shared*. Grouped permutation (e.g. jointly permuting $\{Tas, Hurs, Psl\}$ or all month dummies) would recover combined contributions.

Findings (Fig. 4.1). Both criteria select a consistent hierarchy.

- **First tier:** *Tas* and *Psl* dominate, indicating that, conditional on season, warmer conditions and anticyclonic stability are key determinants of higher $PM_{2.5}$.
- **Second tier:** *Rainfall* (removal by wet scavenging) and `PM25_modelled` (regional background loading) carry substantial signal.
- **Third tier:** *NDVI*, *Hurs*, *SfcWind*, and the *seasonal dummies*, capturing greenness/dispersion, moisture, ventilation, and residual seasonality after conditioning on meteorology. Among months, April and October dummies are the most informative in this ranking.
- **Lower tier:** *PopDen* and `nearest_road_distance` contribute comparatively little at the *monthly, fixed-site* resolution, consistent with mesoscale meteorology and background dominating a single proximity metric.

Synthesis with LM. RF rankings corroborate LM signs: removal/dispersion (*Rainfall*, *SfcWind*) ↓; stagnation/moisture (*Psl*, *Hurs*) ↑; background and greenness modulate the baseline; seasonality persists. RF further captures nonlinearity/interactions, matching its superior hold-out accuracy.



(a) Impurity-based (b) Permutation-based

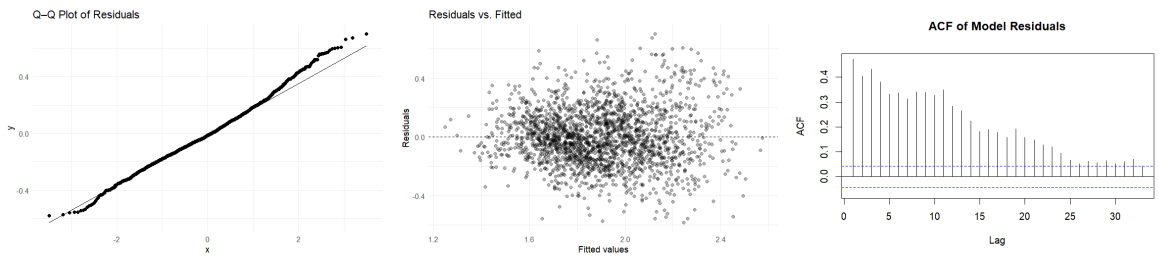
Figure 4.1: Random Forest variable importance on the log target Z. Concordance across criteria indicates a stable ranking of drivers.

Model checking

We report detailed assumption checks for the LM; RF/QRF and conformal procedures require much weaker distributional assumptions for predictive *coverage* and therefore are not subject to the same diagnostics.

LM diagnostics

Distributional shape. The residual Q–Q plot shows central normality with heavy tails at the extremes. To protect coefficient inference, we report HC3 and site–cluster–robust standard errors rather than classical OLS SEs [3, 19, 35].
Homoscedasticity. Residuals vs. fitted indicate mild variance inflation; the studentised Breusch–Pagan test is strongly significant (BP $= 340.9$, $p < 10^{-16}$) [6]. Accordingly, all LM inference uses robust/cluster-robust SEs.
Independence. Durbin–Watson $d = 1.05$ with $\hat{\rho}_1 \approx 0.47$ and the ACF reveal positive serial correlation [10, 11]. To avoid optimistic accuracy estimates under dependence, model evaluation is site-aware: LOSO within training and a strict site-level hold-out [24].
Linearity. Some curvature and interaction structure remains in residual patterns; this helps explain the superior predictive performance of RF, which can accommodate nonlinearity and interactions. The LM is retained as an interpretable baseline selected by AIC, providing a transparent fit–complexity trade-off [1].



(a) Q–Q plot (b) Residuals vs. fitted (c) ACF of residuals

Figure 4.2: LM diagnostics. Heavy tails and heteroskedasticity motivate HC3/cluster-robust SEs; positive autocorrelation motivates site-aware evaluation and distribution-free prediction intervals.

Implication for uncertainty. Because residual normality/homoskedasticity/independence are imperfect, we *do not* rely on parametric PI formulae. Predictive uncertainty is assessed via split-/CV-conformal prediction intervals, which provide finite-sample, distribution-free marginal coverage, and via QRF for heteroskedastic conditional intervals [2, 16, 21].

Sensitivity analyses

We probed robustness to modelling and calibration choices.

(S1) Collinearity handling. Further pruning of highly correlated predictors (beyond `Sun`/`Northing`) and GVIF rechecks left hold-out metrics essentially unchanged: ΔRMSE ≤ 0.03, $\Delta R^2 \leq 0.01$, ΔWidth ≤ 0.05 (μg/m^3). RF/QRF top drivers were invariant.

(S2) LASSO penalty. Using $\lambda_{\min}$ (denser model) slightly reduced LOSO error but altered hold-out results negligibly (RMSE $|\Delta| \leq 0.02$, $R^2|\Delta| \leq 0.01$). We therefore report the more stable $\lambda_{1\text{se}}$.

(S3) Conformal folds. Changing CV-conformal partitions moved coverage by at most $\pm$(1–2) pp and mean PI width by ≤ 0.10 μg/m^3; RF intervals remained narrower than LM/LASSO at comparable coverage.

(S4) RF stochasticity. Across 20 seeds (fixed hyperparameters), hold-out dispersion was modest (RMSE s.d. ≈ 0.02, R^2 s.d. ≈ 0.008). Permutation-importance rankings were stable (median Spearman $\rho > 0.90$); {Tas, Psl, Rainfall, `PM25_modelled`} were always top-4.

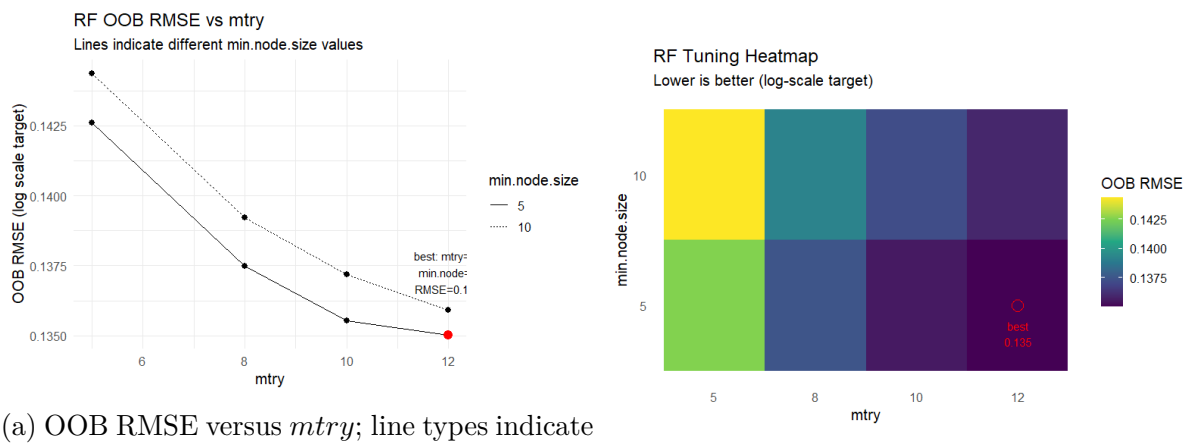
Key implication. Across perturbations, RF/QRF remains best at unseen sites; uncertainty conclusions and scientific interpretation are unchanged.

RF tuning and reproducibility

Grid & criterion. We tuned ($mtry$, `min.node.size`) $\in \{5, 8, 10, 12\} \times \{5, 10\}$ by out-of-bag (OOB) RMSE on the log target Z. The number of trees B was set large enough for the OOB error to stabilise.
Outcome. Both panels of Fig. 4.3 show a monotone decrease in OOB RMSE as $mtry$ increases, with smaller leaf sizes (`min.node.size`= 5) uniformly preferable. The best configuration is

$$mtry = 12, \quad \texttt{min.node.size} = 5, \quad \text{OOB RMSE}_{\log} = 0.135,$$

with diminishing gains beyond $mtry \approx 10$.
Protocol & reproducibility. We fix ($mtry$, `min.node.size`, B) at the OOB–optimal values for all subsequent RF (means) and QRF (quantiles) fits to isolate the loss function from hyperparameters and ensure comparability. Random seeds and software versions are recorded for exact reproducibility.



(a) OOB RMSE versus $mtry$; line types indicate `min.node.size`.

(b) OOB RMSE heatmap (lower is better).

Figure 4.3: RF hyperparameter search on the log target Z. The optimum is at $mtry$=12, `min.node.size`=5; trends are consistent across views.

Chapter 5

Conclusions and Discussion

Summary of conclusions to all questions of interest

What we did

We modelled monthly site-level $PM_{2.5}$ on the log scale with a stepwise-AIC linear model (LM), LASSO, and a Random Forest (RF) with Quantile RF (QRF), back-transforming predictions via Duan smearing [9]. Validation was deployment-oriented: LOSO within training plus a strict site-level hold-out [24]. Uncertainty was assessed using split-/CV-conformal prediction intervals (distribution-free) and QRF conditional quantiles [2, 16, 21, 25, 34].

Main empirical results and interpretation

On the strict site hold-out, **RF/QRF** performed best (RMSE = 1.43, MAE = 0.963, $R^2 = 0.658$) and yielded the narrowest prediction intervals, consistent with tree ensembles capturing nonlinearities and interactions missed by linear baselines [5, 21]. Split-/CV-conformal intervals moved empirical coverage toward the nominal 95% with modest width increases, providing portable, distribution-free uncertainty across learners [2, 16, 34]. Mechanistically, RF importance aligns with LM partial effects: Rainfall and SfcWind $\downarrow$ (removal/ventilation), Psl and Hurs $\uparrow$ (stagnation/moisture), Tas turns positive after conditioning (suppression), `PM25_modelled` $\uparrow$, and NDVI $\downarrow$.

Answers to the two questions of interest

Q1. What is the best model for predicting monthly mean $PM_{2.5}$? *Answer:* The RF/QRF is the best-performing learner under deployment-style evaluation (LOSO + strict site hold-out), providing both accurate point predictions and efficient uncertainty.

Q2. With how much accuracy can monthly means be predicted? *Answer:* On the strict site hold-out the achievable accuracy is RMSE ≈ 1.43 μg/m^3 and MAE ≈ 0.96 μg/m^3 with $R^2 \approx 0.66$. Conformal PIs attain near-nominal coverage with mean widths ≈ 4.5 μg/m^3 for RF after CV calibration.

Imputing missing $PM_{2.5}$ with the tuned QRF

Using the hold-out winner, we imputed missing values with the tuned QRF median on the log target and back-transformed to the original scale. Pre-processing (standardisation and design-matrix alignment) exactly mirrored training. In total we **imputed** 1,893 **of** 4,560 **rows**; the file `Data - PM25_imputed.csv` contains the point fill (`PM25_filled`) and 2.5%/97.5% predictive limits.

Strengths and limitations

Strengths

Four features strengthen this study. (i) *Deployment-aligned validation*—LOSO plus a strict site-level hold-out—targets spatial generalisation and curbs optimism [24]. (ii) *Distribution-free uncertainty*: split/CV-conformal intervals provide finite-sample marginal coverage and transfer across learners [16, 34]. (iii) *Dual interpretability*: LM partial effects and RF importance tell a consistent physical story (removal, dispersion, stagnation, background, greenness). (iv) *Reproducibility*: fixed folds/seeds, recorded hyperparameters/software, and a hand-off imputed dataset.

Limitations

Key limitations remain. (i) *Resolution/representativeness*: monthly aggregation and a fixed network miss sub-monthly and microscale signals; extrapolation to new regimes is uncertain. (ii) *Covariate quality/confounding*: background, land-use and satellite proxies are coarse/noisy, so residual confounding may persist. (iii) *Dependence/drift*: temporal autocorrelation and non-stationarity can erode performance in operation, calling for drift monitoring and recalibration [14]. For missing responses, we used predictive *single* imputation (QRF median); *multiple* imputation is preferable for inference [18, 33].

Concluding remarks and future work

Under site-aware validation, RF/QRF generalises best and paired with conformal calibration delivers reliable uncertainty; the fitted relationships accord with physical dispersion, removal, and stagnation mechanisms. The released imputed dataset fills 1,893 gaps with predictive bands while preserving distributions and seasonality, yielding a deployable pipeline for monthly $PM_{2.5}$ in Scotland. Looking ahead, we will operationalise *drift-aware deployment* (rolling revalidation, scheduled recalibration, and change detection) to guard against concept drift [14]; refine uncertainty via *groupwise conformal calibration* and *conformalized quantile regression* for conditional coverage [2, 25]; deepen interpretability with *grouped permutation importance* for correlated blocks and SHAP-based local attributions [20]; and replace single imputation with *multiple imputation* (e.g., MICE) to propagate imputation uncertainty in downstream analyses [18, 33].

References

[1] Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723.

[2] Angelopoulos, A. N. and Bates, S. (2023). A gentle introduction to conformal prediction and distribution-free uncertainty quantification. Pedagogical survey.

[3] Arellano, M. (1987). Computing robust standard errors for within-groups estimators. *Oxford Bulletin of Economics and Statistics*, 49(4):431–434.

[4] Box, G. E. P. and Cox, D. R. (1964). An analysis of transformations. *Journal of the Royal Statistical Society: Series B*, 26(2):211–252.

[5] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.

[6] Breusch, T. S. and Pagan, A. R. (1979). A simple test for heteroscedasticity and random coefficient variation. *Econometrica*, 47(5):1287–1294.

[7] Carroll, R. J. and Ruppert, D. (1988). *Transformation and Weighting in Regression*. Chapman and Hall, New York.

[8] Department for Environment, Food & Rural Affairs and the Devolved Administrations (2024). Air pollution in the UK 2023. Technical report, DEFRA, London.

[9] Duan, N. (1983). Smearing estimate: A nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383):605–610.

[10] Durbin, J. and Watson, G. S. (1950). Testing for serial correlation in least squares regression. I. *Biometrika*, 37(3-4):409–428.

[11] Durbin, J. and Watson, G. S. (1951). Testing for serial correlation in least squares regression. II. *Biometrika*, 38(1-2):159–178.

[12] Fox, J. and Weisberg, S. (2019). *An R Companion to Applied Regression*. Sage, 3rd edition.

[13] Friedman, J., Hastie, T., and Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1):1–22.

[14] Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., and Bouchachia, A. (2014). A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4):44:1–44:37.

[15] Health Effects Institute (2020). State of global air 2020: Special report. Technical report, Health Effects Institute, Boston, MA. Reports that in 2019 air pollution contributed to 6.67 million deaths worldwide; ambient $PM_{2.5}$ contributed to 4.14 million deaths.

[16] Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R., and Wasserman, L. (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111.

[17] Limpert, E., Stahel, W. A., and Abbt, M. (2001). Log-normal distributions across the sciences: Keys and clues. *BioScience*, 51(5):341–352.

[18] Little, R. J. A. and Rubin, D. B. (2019). *Statistical Analysis with Missing Data*. Wiley, 3 edition.

[19] Long, J. S. and Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, 54(3):217–224.

[20] Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*.

[21] Meinshausen, N. (2006). Quantile regression forests. *Journal of Machine Learning Research*, 7:983–999.

[22] Osborne, J. W. (2010). Improving your data transformations: Applying the box–cox transformation. *Practical Assessment, Research, and Evaluation*, 15(12):1–9.

[23] Ott, W. R. (1990). A physical explanation of the lognormality of pollutant concentrations. *Journal of the Air & Waste Management Association*, 40(10):1378–1383.

[24] Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D., Wintle, B., Hartig, F., and Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929.

[25] Romano, Y., Patterson, E., and Candès, E. J. (2019). Conformalized quantile regression. In *Advances in Neural Information Processing Systems (NeurIPS)*.

[26] Rouse, J. W., Haas, R. H., Schell, J. A., and Deering, D. W. (1974). Monitoring vegetation systems in the great plains with ERTS. In *Third Earth Resources Technology Satellite–1 Symposium*, volume 1, pages 309–317, Washington, D.C. NASA.

[27] Scottish Government (2021). Cleaner air for scotland 2: Towards a better place for everyone.

[28] Strobl, C., Boulesteix, A.-L., Kneib, T., Augustin, T., and Zeileis, A. (2008). Conditional variable importance for random forests. *BMC Bioinformatics*, 9:307.

[29] Strobl, C., Boulesteix, A.-L., Zeileis, A., and Hothorn, T. (2007). Bias in random forest variable importance measures: Illustrations, sources and a solution. *BMC Bioinformatics*, 8:25.

[30] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288.

[31] UK AIR (2024). Automatic urban and rural network (AURN). UK Air Information Resource. Overview of the UK's compliance monitoring network.

[32] UK Air Information Resource (UK-AIR) and Department for Environment, Food & Rural Affairs (DEFRA) (2023). Pollution climate mapping (PCM) model: mapped annual background $PM_{2.5}$ concentrations (1 km^2) for the UK. Technical report, UK-AIR/DEFRA. Annual 1 km^2 mapped background $PM_{2.5}$ widely used in UK air-quality assessment.

[33] van Buuren, S. (2018). *Flexible Imputation of Missing Data*. CRC Press, 2 edition.

[34] Vovk, V., Gammerman, A., and Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer, New York, NY.

[35] White, H. (1980). A heteroscedasticity-consistent covariance matrix estimator and a direct test for heteroscedasticity. *Econometrica*, 48(4):817–838.

[36] World Health Organization (2021). WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. Technical report, World Health Organization, Geneva.

[37] Wright, M. N. and Ziegler, A. (2017). ranger: A fast implementation of random forests for high dimensional data in C++ and R. *Journal of Statistical Software*, 77(1):1–17.